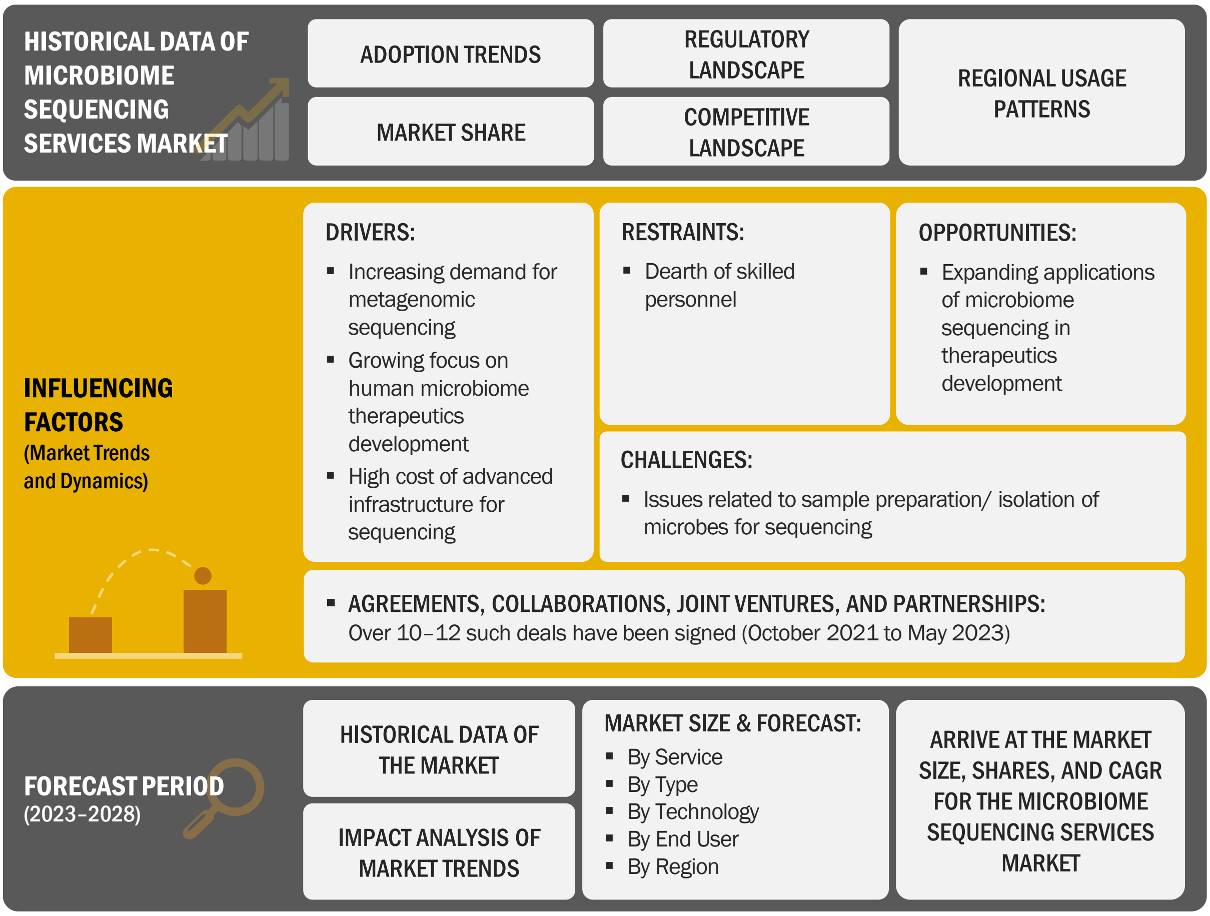
2 RESEARCH METHODOLOGY

2.1 RESEARCH DATA

This market research study involved the extensive use of secondary sources, directories, and databases to identify and collect information useful for this technical, market-oriented, and financial study of the global microbiome sequencing services market. In-depth interviews were conducted with various primary respondents, including key industry participants, subject-matter experts (SMEs), C-level executives of key market players, and industry consultants, to obtain and verify critical qualitative and quantitative information and assess market growth prospects. The global size of the microbiome sequencing services market (estimated through various research approaches) was then triangulated with inputs from primary research to arrive at the final market size.

FIGURE 1 RESEARCH DESIGN



Sources: Secondary Research and MarketsandMarkets Analysis

2.1.1 SECONDARY DATA

The secondary sources referred to for this research study include publications from government sources, such as the National Human Genome Research Institute (NHGRI), the National Center for Biotechnology Information (NCBI), the Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation France, the Centre for Genomic Regulation (CRG) Spain, Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC) Australia, Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome) India, South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis.

Secondary sources include corporate and regulatory filings (such as annual reports, SEC filings, investor presentations, and financial statements); business magazines and research journals; press releases; and trade, business, and professional associations. Secondary data was collected and analyzed to arrive at the overall size of the global microbiome sequencing services market, which was validated through primary research.

Secondary research involved three major activities:

Background Study

- Building a basic understanding of the microbiome sequencing services market
- Analyzing the MarketsandMarkets data repository, followed by drawing and updating data tables using current information
- Identifying data gaps and key global players/service providers in each segment

Market Understanding

- Identifying stakeholders and key decision-makers in different regions
- Identifying the selection criteria of key decision-makers
- Analyzing the competitive landscape
- Analyzing major global players with their respective service portfolios
- Analyzing the strategies adopted by global players to position their services offered
- Analyzing markets at the regional/country level

Trends

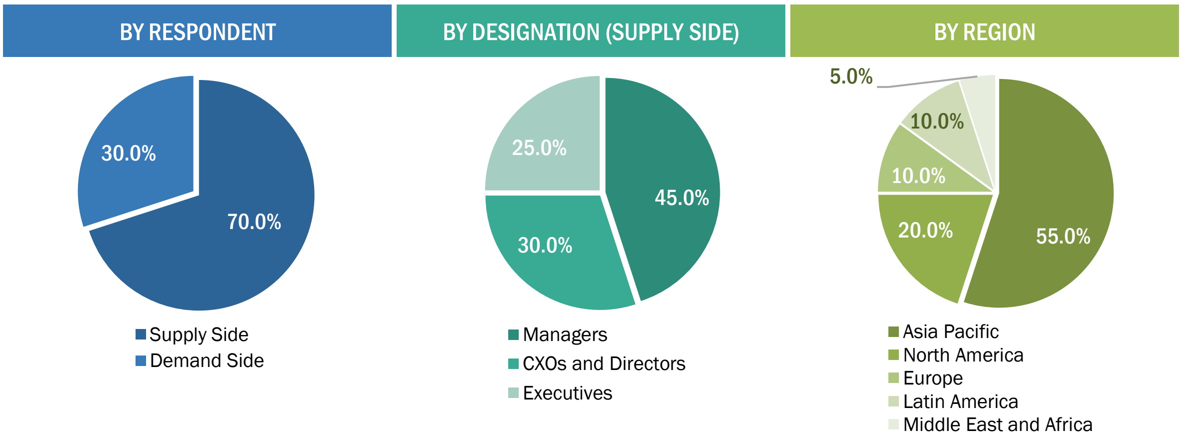
- Identifying and analyzing key products and industry trends to draw notes and include considerations in the final forecasts
- Identifying the major drivers, restraints, opportunities, and challenges in the market
- Analyzing the regulatory landscape and industry trends

2.1.2 PRIMARY DATA

Extensive primary research was conducted after acquiring basic knowledge about the global microbiome sequencing services market through secondary research. Several primary interviews were conducted with market experts from both the demand side (personnel from pharmaceutical and biopharmaceutical companies, medical device companies, and academic institutes) and supply side (C-level and D-level executives, product managers, and marketing and sales managers of key manufacturers, distributors, and channel partners, among others, from tier 1 and tier 2 companies engaged in offering services) across five major regions—North America, Europe, the Asia Pacific, Latin America, and the Middle East & Africa. Approximately 70% and 30% of primary interviews were conducted with supply-side and demand-side participants, respectively. This primary data was collected through questionnaires, e-mails, online surveys, personal interviews, and telephonic interviews.

FIGURE 2 MICROBIOME SEQUENCING SERVICES MARKET: BREAKDOWN OF PRIMARIES

SUPPLY-SIDE PARTICIPANTS		DEMAND-SIDE PARTICIPANTS
Microbiome Insights, Inc.	Rebiotix Inc.	Genefitletics
CosmosID	BaseClear	BIONE
Microba Life Sciences	Clarity Genomics, Inc.	Viome Life Sciences Inc.



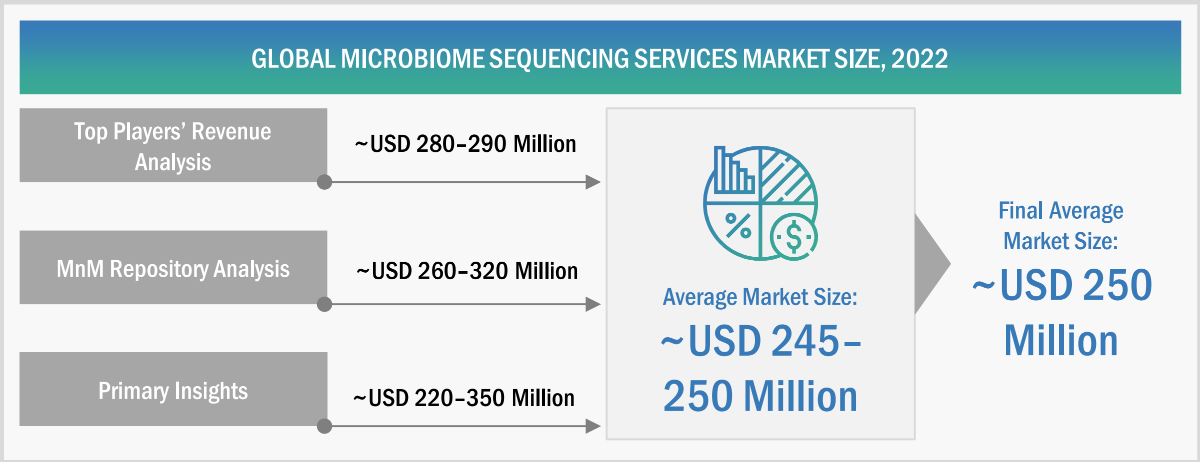
Primary research was used in this report to:

- Validate the market segmentation defined through a service portfolio assessment of the leading players in the microbiome sequencing services market
- Understand key industry trends and issues defining the strategic growth objectives of market players
- Gather both demand and supply-side validation of the key factors affecting the market growth (such as market drivers, restraints, challenges, and opportunities)
- Validate assumptions for the market sizing and forecasting model used for this study
- Understand the market position of leading players in the microbiome sequencing services market and their market shares/ranking

2.2 MARKET ESTIMATION METHODOLOGY

The report presents a detailed assessment of the microbiome sequencing services market and qualitative inputs and insights from MarketsandMarkets. The global size of the microbiome sequencing services market was estimated through multiple approaches. A detailed market estimation approach was followed to estimate and validate the value of the microbiome sequencing services market and other dependent submarkets, as mentioned below:

FIGURE 3 MICROBIOME SEQUENCING SERVICES MARKET SIZE ESTIMATION (SUPPLY-SIDE ANALYSIS), 2022

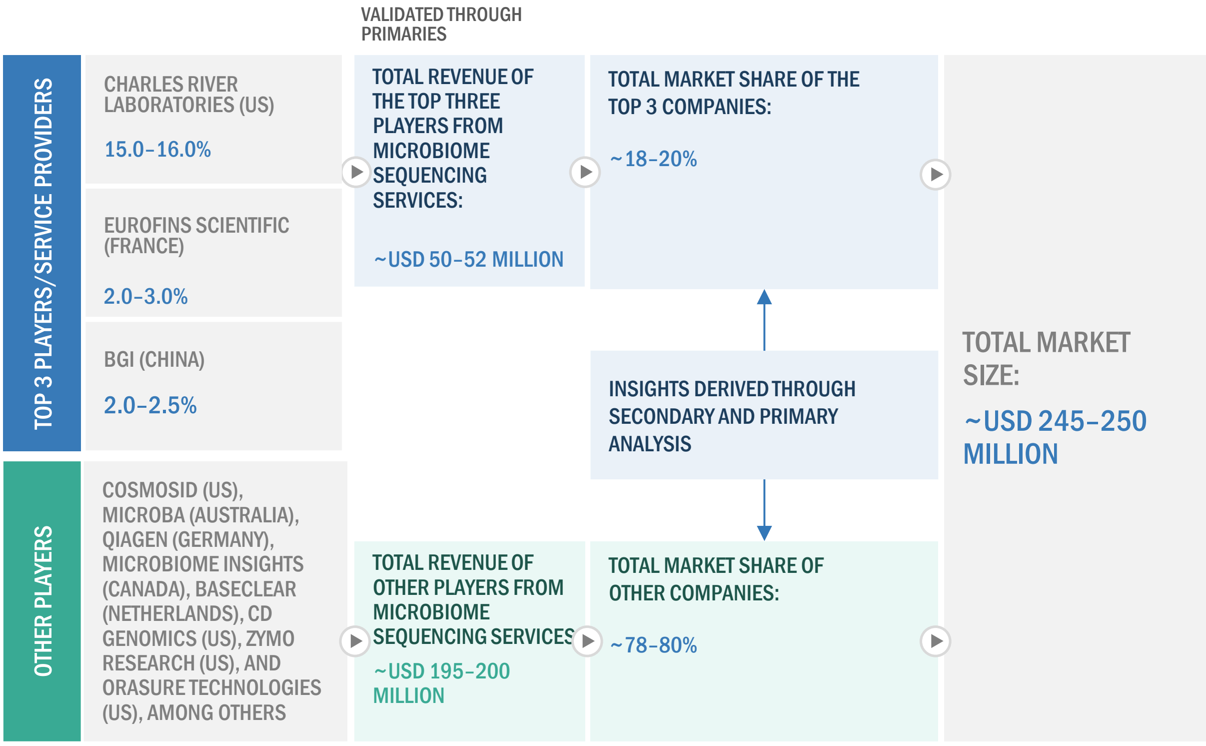


Source: MarketsandMarkets Analysis and Primary Insights

APPROACH 1: COMPANY REVENUE ANALYSIS (BOTTOM-UP APPROACH)

- Revenues for individual companies were gathered from public sources and databases.
- The microbiome sequencing services-related business shares of leading players were gathered from secondary sources to the extent available. In certain cases, the shares of microbiome sequencing services businesses were ascertained through a detailed analysis of various parameters, including service portfolios, market positioning, and primary insights.
- Individual shares or revenue estimates were validated through expert interviews.

FIGURE 4 MARKET SIZE ESTIMATION: APPROACH 1 (COMPANY REVENUE ANALYSIS-BASED ESTIMATION), 2022

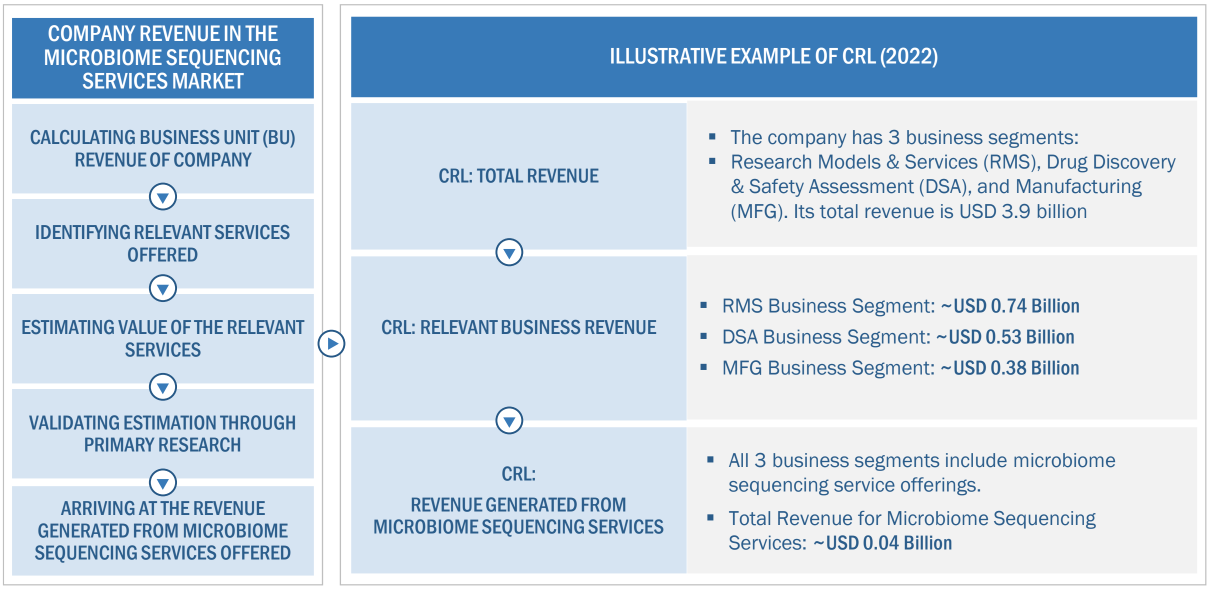


Source: MarketsandMarkets Analysis and Primary Insights

For the calculation of the global market value, the segmental revenue was arrived at based on the revenue mapping of major players active in the microbiome sequencing services market. This process involved the following steps:

- A list of the major global players operating in the microbiome sequencing services market was generated.
- The annual revenues generated by major global players from the microbiome sequencing services market, or the nearest reported business unit/category, were mapped.
- The revenues of the top 3 players in the microbiome sequencing services market were identified.
- The revenue of the top 3 players was assumed to be approximately 18–20% in 2022 and added up to 100% to derive the global value of the microbiome sequencing services market.

FIGURE 5 ILLUSTRATIVE EXAMPLE OF COMPANY REVENUE ANALYSIS, 2022



Source: Annual Reports, SEC Filings, Investor Presentations, Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

APPROACH 2: MNM REPOSITORY ANALYSIS

- For estimating the microbiome sequencing services market, MnM reports, such as the contract research organization (CRO) services market, next-generation sequencing (NGS) market, and metagenomic sequencing market, were considered. The global and regional market values of microbiome sequencing services and the dependent submarkets were extracted from the MnM repository and validated through secondary and primary research. The final global market size was triangulated through the average of all approaches and validated through primary interviews with industry experts.

APPROACH 3: SECONDARY RESEARCH

- The size of the microbiome sequencing services market was obtained from secondary sources.
- The shares of various microbiome sequencing services in the overall market were obtained from secondary data and validated through primary participants to arrive at the total microbiome sequencing services market.
- Primary participants further validated the numbers.

APPROACH 4: PRIMARY INTERVIEWS

- As part of the primary research process, individual respondents were interviewed for insights on market size and market growth.
- All responses were collated, and a weighted average was taken to derive a probabilistic estimate of the market size and growth rate.

2.2.1 INSIGHTS FROM PRIMARY EXPERTS

FIGURE 6 MARKET SIZE VALIDATION FROM PRIMARY SOURCES

FACTORS AFFECTING MARKET GROWTH

Product Manager	The current market is growing bigger in size as new players are entering the market with varied services for microbiome sequencing.
Group Leader	The key sequencing methods popular for microbiome profiling include amplicon sequencing and shotgun sequencing.
Research Scientist	Emerging technologies such as nanopore sequencing are set to provide lucrative opportunities for market growth.

MARKET TRENDS

R&D investments in oncology applications are positively impacting the market.

Growing adoption of new sequencing tools is expected to drive service adoption.

– Market Experts

ACCORDING TO A MAJORITY OF INDUSTRY EXPERTS, THE TOP MARKET PLAYERS ACCOUNTED FOR ~20% OF THE MICROBIOME SEQUENCING SERVICES MARKET

COMPANY	MARKET RANK (2022)
CRL (US)	1
Eurofins Scientific (France)	2
BGI (China)	3

Source: Primary Insights

The final market size was estimated based on the weighted average of values arrived at through various approaches, such as company revenue analysis, MnM repository analysis, secondary research, and primary insights. The research methodology and market numbers were further validated from primaries.

SEGMENT ASSESSMENT (BY SERVICE, TYPE, TECHNOLOGY, AND END USER)

The segmental assessment was conducted using the following approaches:

- Approach for service: Revenue contributions/splits of microbiome sequencing services were calculated from the total revenues of leading players in the market (wherever available), and respective growth trends were derived by studying the adoption patterns of various services considered in the study.
- Approach for type: Revenue contributions/splits of different segments (amplicon sequencing, shotgun sequencing, and whole-genome sequencing) to the total revenues of leading players in the market (wherever available) and respective growth trends were derived by studying the adoption patterns of various products considered in the study.
- Approach for technology: Splits for segments were derived by studying the adoption preference (based on the type of service, sample size, and turnaround time) for different technologies used in services, including sequencing by synthesis (SBS), sequencing by ligation (SBL), and nanopore sequencing. This information was validated through primary interviews.
- Approach for end users: Splits for segments were derived by studying the adoption patterns of microbiome sequencing services by different end users in the market.

- Approach for regions: The geographic assessment was conducted by studying geographic revenue contributions/splits of leading players in the market (wherever available) and respective growth trends.

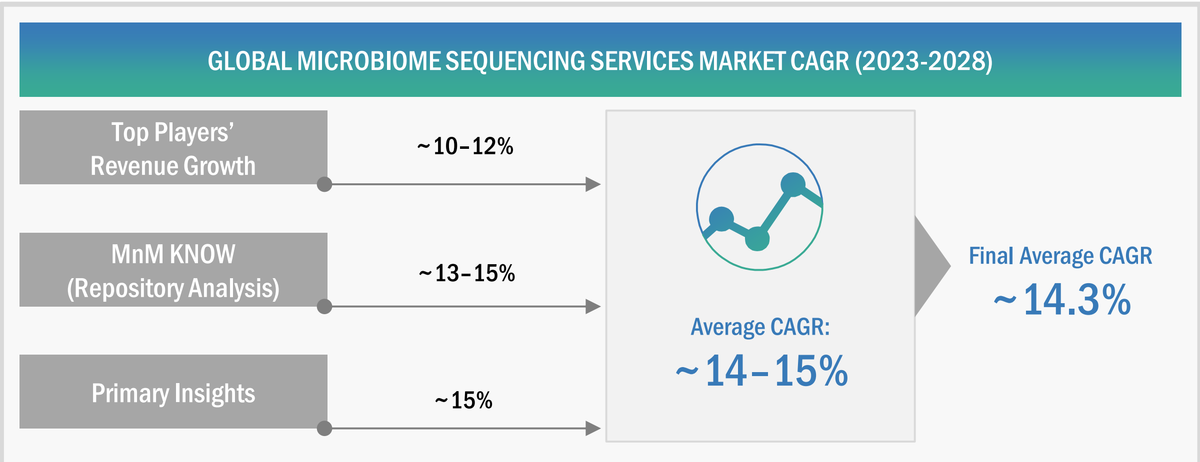
The assumptions and approaches were validated at each point through industry experts contacted during primary research. Considering the limitations of data available from secondary research, revenue estimates for individual companies (for the overall microbiome sequencing services market and geographic market assessment) were ascertained based on a detailed analysis of their respective service offerings, geographic reach/strength (direct or through distributors or suppliers), and shares of the leading players in a particular region or country.

2.3 MARKET GROWTH RATE PROJECTIONS

The growth forecast model was defined based on an assessment of the qualitative and quantitative factors affecting the market growth. These mainly included:

1. Historic revenue growth trends of the leading players in the market
   - Demand-side factor assessment, including clinical trials conducted for microbiome therapeutics, pharmaceutical R&D spending, the incidence of various diseases, the number of drug discovery and development projects across all geographies, and trial density, among other factors
   - Impact assessment of the key factors, such as drivers, restraints, challenges, and opportunities, influencing market growth
   - Impact assessment of the historic competitive trends driving market growth—service launches, strategic growth initiatives, innovation trends, and R&D initiatives
   - Impact assessment of regulatory guidelines/mandates on market growth over the forecast period. Only the defined regulatory guidelines related to the study period were considered for this assessment. The impact of any future regulatory uncertainties on market growth is not accounted for.
   - The impact of the recession on the overall market was evaluated for 2023–2028

FIGURE 7 MICROBIOME SEQUENCING SERVICES MARKET (SUPPLY-SIDE): CAGR PROJECTIONS



Source: MarketsandMarkets Analysis and Primary Insights

Considering the average of all the growth rates from the approaches mentioned above, the final CAGR for 2023–2028 was estimated to be ~14.3% for this report.

FIGURE 8 MICROBIOME SEQUENCING SERVICES MARKET: GROWTH ANALYSIS OF DEMAND-SIDE DRIVERS

DRIVERS	IMPACT LEVEL
Increasing demand for metagenomic sequencing	●
Growing focus on human microbiome therapeutics development	●
High cost of advanced infrastructure for sequencing	●
RESTRAINTS	**IMPACT LEVEL**
Dearth of skilled personnel	●
OPPORTUNITIES	**IMPACT LEVEL**
Expanding applications of microbiome sequencing in therapeutics development	●
CHALLENGES	**IMPACT LEVEL**
Issues related to sample preparation/isolation of microbes for sequencing	●

IMPACT LEVEL: ● High ● Medium ● Low

Source: Secondary Research, MarketsandMarkets Analysis, and Primary Insights

2.4 MARKET BREAKDOWN AND DATA TRIANGULATION

After arriving at the market size from the market size estimation process explained above, the total market was divided into several segments. To complete the overall market engineering process and arrive at the exact statistics for all segments, data triangulation and market breakdown procedures were employed, wherever applicable.

The following figure shows the market validation sources structure and the data triangulation methodology implemented in the market engineering process for making this report.

FIGURE 9 DATA TRIANGULATION METHODOLOGY

PRIMARY SOURCES

DEMAND SIDE

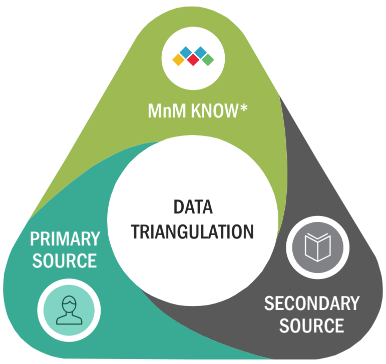
INTERVIEWS WITH PERSONNEL FROM:

- Contract Research Organizations
- Pharmaceutical and Biopharmaceutical Companies
- Medical Device Manufacturing Companies
- Academic Institutes

SUPPLY SIDE

INTERVIEWS WITH:

- CEOs
- Presidents
- Vice Presidents
- Marketing Managers
- Business Managers
- Sales Executives
- Other Microbiome Sequencing Services Market-related Personnel



SECONDARY SOURCES

Annual Reports, Presentations, Websites, and Press Releases of Top Players; News Articles; Journals; and Paid Databases

- National Human Genome Research Institute (NHGRI)
- National Center for Biotechnology Information (NCBI)
- The Genomics Research and Development Initiative (GRDI)
- UK Research and Innovation (UKRI)
- Microbiome Foundation France
- The Centre for Genomic Regulation (CRG), Spain
- Human MetaGenome Consortium Japan (HMGJ)
- Microbiome Research Centre (MRC), Australia
- Center of Innovation in Personalized Medicine (CIPM)
- Department of Biotechnology (GENOME India and Microbiome), India
- South African Society for Human Genetics
- Food and Drug Administration (FDA)

INFORMATION SOURCED

Key Players	Competitive Landscape	Opportunities/ Challenges	Drivers and Restraints

Market Size (2022)	Forecast (2023–2028)	Geographical Analysis

MnM KNOW* stands for MarketsandMarkets' 'Knowledge Asset Management' framework. In this context, it stands for existing market research knowledge repository of over 5,000 granular markets, our flagship competitive intelligence and market research platform "Knowledge Store," subject matter experts, and independent consultants. MnM KNOW acts as an independent source that helps us validate information gathered from primary and secondary sources.

2.5 RESEARCH ASSUMPTIONS

- Market values have been rounded off to the nearest decimal. Due to this, there may be slight differences between actual totals and the totals mentioned in market tables.
- Region-wise inflation/deflation for service pricing during 2023–2028 has not been considered.
- It has been assumed that dollar fluctuations will not affect the growth forecast to a significant extent.
- Revenues of private companies acquired from paid databases were assumed to be in line with actual company financials.
- A positive economic climate will continue in emerging countries until 2028, and the recession of 2023 will have a low impact on the microbiome sequencing services market.
- There will be no substantial changes in government regulations that will significantly impact the market.
- All market segments and subsegments listed in this report are considered to be mutually exclusive.
- Wherever the exact segmental market splits were not available, historical trends and inputs from primary respondents have been considered.

2.6 RISK ANALYSIS

DESCRIPTION	RISK ANALYSIS
The growth rate for each segment is considered based on recent market trends, key developments of companies, analysis of historical growth rates, and primary insights.	Moderate
Market sizes and shares for 2022 were estimated based on the financials available for nine months in company annual/quarterly reports and presentations for a few companies.	Low to Moderate

2.7 IMPACT OF RECESSION ON MICROBIOME SEQUENCING SERVICES MARKET

A recession is a period of temporary economic decline during which trade and industrial activities are reduced. It is generally identified by a decline in the GDP in two successive quarters. According to the World Economic Forum Forecast Report (October 2022), global GDP growth would slow from 6% in 2021 to 3.2% in 2022 and 2.7% in 2023. Alternatively, the same report suggests that global inflation would rise from 4.7% in 2021 to 8.8% in 2022 but decline to 6.5% in 2023 and to 4.1% by 2024.

The key reasons behind this slowdown are the Russia-Ukraine war (which led to a severe energy crisis in Europe, where gas prices have escalated four-fold since 2021) and frequent lockdowns in China under its 'Zero COVID policy,' which impacted global trade and led to a sharp appreciation of the US dollar.

Based on analysis by the pharmaceutical and biotechnology industry and inputs from primary experts, the pharmaceutical and biotechnology industry is expected to withstand this downturn. The demand for healthcare products and services is expected to endure, and the recession will have a low to minimal impact on overall healthcare-oriented businesses.